Janhavi Chaurasia

Email: janhavic@engineering.upenn.edu GitHub: https://github.com/JanhaviChaurasia

EDUCATION

Bachelors of Technology in Computer Science Engineering **[Sep'22 - Apr'26]**
School of Computer Science at Vellore Institute of Technology Chennai Campus
CGPA: 9.14/10 [3.901/4]

- Coursework: Cognitive Robotics, Machine Vision, Artificial Intelligence, Natural Language Processing, Linear Algebra, Differential Equations, Discrete Math and Graph Theory, Data Structures and Algorithms

SKILLS

Language and Framework : C/C++, Python, ROS/ROS2, OpenCV, Arduino, TensorFlow, Pytorch, OpenAI-GYM, MAVLink
Simulation and Modelling : Gazebo, MATLAB, Rviz, IsaacLab/IsaacSim; **Version Control and Containerization** : Git, Docker

RESEARCH & PUBLICATIONS

Dopamine In The Mitigation Of Catastrophic Forgetting In Reinforcement Learning
Published at 2025 5th International Conference on Machine Learning and Intelligent Systems Engineering [Paper] [GitHub]

- Developed dopamine-inspired reinforcement learning methods integrating Elastic Weight Consolidation, neuromodulated temporal difference updates, and Hebbian learning to stabilize DQNs across sequential Atari tasks.
- Introduced dopamine-gated synaptic consolidation achieving near-zero forgetting (Backward Transfer = 0) and >99% reduction in weight changes compared to baseline DQN.
- Achieved positive backward transfer (+10) with neuromodulated learning rate, indicating reinforcement of prior knowledge during new task learning.

SwarmFormer: Transformer-Based Multi-Robot Formation Control With Risk-Aware Fusion
Published at 8th International Conference on Robotics, Control and Automation Engineering 2025 [GitHub][Dataset][Paper]

- Developed SwarmFormer, a goal-conditioned transformer policy utilizing hierarchical cross-modal attention and Gaussian Mixture Model outputs for robust decentralized multi-robot formation control.
- Fused diverse input modalities including odometry, CVaR-augmented LiDAR, explicit formation goals, and decentralized gossip communication via an asynchronous scalable protocol.
- Created and trained on a comprehensive Gazebo-ROS2 dataset covering 5 formation types, 4 swarm sizes, multiple obstacle densities, and varied communication topologies.
- Achieved a test Mean Absolute Error (MAE) of 0.0767, improving 21% over MLP and 28% over RNN baselines.
- Demonstrated strong probabilistic modeling with Negative Log-Likelihood (NLL) of -3.67, enabling uncertainty-aware control in complex environments.

SGSPose: Neuromorphic-Geometric 6D Pose Estimation through Spiking Graph Neural Networks & SE3-Equivariant Learning
Published at IEEE Access [Paper][GitHub]

- Developed SGSPose, a novel 6D pose estimation architecture integrating spiking neural networks for temporal spike encoding with SE(3)-equivariant graph neural networks enforcing geometric consistency for robust pose regression.
- Designed an elegant pipeline converting RGB images into temporal spiking features, followed by SE(3)-equivariant graph message passing with orthogonalized rotation outputs for precise, physically consistent 6D pose estimation.
- Utilized a composite loss combining position error (Smooth L1), geodesic rotation distance, and ADD-S metric with adaptive weighting, optimizing translation and rotation simultaneously.
- Achieved over 90% improvement in median translation error on 7Scenes compared to PoseNet, with translation errors as low as 0.019m and average below 0.06m; rotation errors range from 8.3° to 21.3° with ADD-S accuracy up to 96.2%.
- Conducted as part of a research internship at Centre for Cyber Physical Systems (CCPS) at VIT Chennai.

Autonomous Exploration Of Unknown Indoor Environments Using Multi-UAV System
Published at 2025 5th International Conference on Computer, Control and Robotics [Paper] [GazeboSim] [Pygame]

- Developed a coordinated multi-UAV swarm system for autonomous indoor exploration using a master-slave architecture with dynamic task allocation and safety-aware coordination via ROS2.
- Integrated PX4 autopilot, Gazebo simulator, and ROS2 middleware for high-fidelity UAV simulation, sensor fusion, and GPS-denied navigation using LiDAR-based SLAM and Visual-Inertial Odometry.
- Designed master and slave drone algorithms for intelligent floor assignment, adaptive swarm formation, recursive frontier exploration, obstacle detection, and battery-aware autonomous exploration and safe return.

EXPERIENCE

Robotics Intern, Accio Robotics, Bengaluru **[Jan'26 - Jun'26]**
Bin Picking and Multi-Robot Coordination System Development

- Conducted research on explainable robotic diagnostics, leading the development of NS-DART, a neuro-symbolic framework combining graph neural networks, temporal representation learning, physics-informed reasoning, and

LLM-based explanations for autonomous robotic systems.

- Audited and extended production robotic software frameworks to improve system reliability, maintainability, and integration across existing robot platforms.

Robotics Intern, Accio Robotics, Bengaluru **[May’25 - Jul’25]**

Bin Picking and Multi-Robot Coordination System Development

- Developed an industrial bin picking pipeline using YOLOv8 few-shot instance segmentation on RGBD data, chosen after evaluating Mask R-CNN and vision-language models to balance real-time performance with budget and compute limits.
- Built an interactive annotation tool with SAM2 integration, providing polygon and magnetic lasso editing, undo-redo, and YOLO export to accelerate high-quality dataset creation.
- Engineered projection homography pipelines with RealSense sensors for precise alignment and spatial consistency.

Robotics Intern, Clearspot.ai **[Aug’24 - Dec’24]**

UAV Simulation and Perception System Development

- Processed and integrated image metadata for precise georeferencing and mapping within the software framework.
- Developed a UAV simulation environment using PX4 and ROS2, customizing flight dynamics and control algorithms to meet project specifications.
- Implemented computer vision techniques with OpenCV to analyze real-time drone camera feeds, enabling feature extraction and validation of perception-based navigation.

SELECTED PROJECTS

Quadruped Gait Imitation Learning with Sensorimotor Feedback **[Apr’25 – Aug’25]**

Sensorimotor Feedback Control for Legged Robots using Imitation Learning and Predictive Modeling [GitHub]

- Designed and tuned a biologically inspired Central Pattern Generator (CPG) controller, collecting 2,075 high-fidelity demonstrations with explicit phase info for imitation learning of robust quadruped gaits.
- Developed behavioral cloning using multi-layer perceptrons on 18-dimensional inputs (16 joint angles + 2 phase features), achieving a forward model RMSE of 0.00185 radians (~0.11°), ensuring stable trotting gait reproduction.
- Engineered a deep forward predictive model with sensorimotor feedback control to detect gait anomalies and apply corrections in real-time, improving forward progress recovery by 399% under perturbations without loss of stability.

Multi-Agent Reinforcement Learning for Swarm Pursuit Coordination **[Nov’24 - Mar’25]**

Pac-Man Ghosts as Multi-Agent Reinforcement Learning Swarm Pursuers [GitHub]

- Developed multi-agent RL framework implementing IQL, QMIX, and MADDPG within a custom Pac-Man simulation representing a multi-robot swarm pursuit problem.
- Applied a team-oriented reward function to encourage cooperative behavior, stabilize learning, and handle scalability and non-stationarity challenges.
- Achieved 100% and 90% win rates for QMIX and MADDPG respectively, with average episode lengths converging to ~48 steps; IQL lagged with 70% win rate and longer episodes (~73 steps).
- MADDPG delivered the highest average rewards (~186) demonstrating the most stable performance across 500 episodes.

Multi-TurtleBot Coordination in Gazebo **[Jul’24 - Oct’24]**

Multi-TurtleBot Navigation for Target Search and Goal Achievement [Github]

- Customized Gazebo and integrated ROS2 for spawning and coordinating multiple TurtleBots in a shared environment.
- Implemented the Bug Algorithm for autonomous wall-following, enabling collaborative target search.
- Applied OpenCV for real-time target detection and localization during search.
- Designed navigation strategies for autonomous goal-directed movement upon detection.

Perception for Autonomous Underwater Vehicle **[Jan’24-May’24]**

Perception system for autonomous underwater vehicles with ArUco pose estimation [Github]

- Developed a comprehensive underwater perception and navigation stack integrating ArUco-based pose estimation in ROS for accurate localization and mapping in dynamic environments.
- Integrated camera sensors within waterproof housing to ensure robust data acquisition and processing underwater.
- Laid the foundation for autonomous vehicle decision-making enabling precise navigation and task execution.

AWARDS & LEADERSHIP

- **TAC Challenge Runners-Up & Team Leadership, Dreadnought Robotics, VIT Chennai** **[Sep'23 - Aug'25]**
  As Head of Operations (2023-24), led the team to global Runners-Up at TAC Challenge 2024, Norway by overseeing coordination, technical and financial decisions, and developing ArUco marker-based perception and navigation stack for AUV. Advanced to Team Manager (2024-25), managing 120+ students, conducting component research and development methodology evaluation, overseeing finances and operations, and mentoring across robotics domains.
- **Management Team Lead, Google Developers Students’ Club, VIT Chennai** **[Sep’23 - Sep’24]**
  Directed team recruitment, delegation, and event management; implemented financial planning and produced technical documentation to ensure smooth execution of diverse technical events.

LINKS

Research Links

- Dopamine In The Mitigation Of Catastrophic Forgetting In Reinforcement Learning
    - Paper : https://ieeexplore.ieee.org/abstract/document/11100173
    - GitHub : https://github.com/JanhaviChaurasia/DopamineReinforcementLearning
- SwarmFormer: Transformer-Based Multi-Robot Formation Control With Risk-Aware Fusion
    - Paper : https://drive.google.com/file/d/1X7AYw-P2fILt9LUmRwFreE6ufEVuHRJ3/view?usp=sharing
    - GitHub : https://github.com/JanhaviChaurasia/SwarmFormer
    - Dataset : https://github.com/JanhaviChaurasia/SwarmDataset
- SGSPose: Neuromorphic-Geometric 6D Pose Estimation through Spiking Graph Neural Networks & SE3-Equivariant Learning
    - Paper : https://ieeexplore.ieee.org/document/11338761
    - GitHub : https://github.com/JanhaviChaurasia/SGSPose
- Autonomous Exploration Of Unknown Indoor Environments Using Multi-UAV System
    - Paper : https://ieeexplore.ieee.org/abstract/document/11072559
    - Pygame : https://github.com/JanhaviChaurasia/PygameDroneSwarm
    - GazeboSim : https://github.com/JanhaviChaurasia/DroneSwarmNavigation

Project Links

- Quadruped Gait Imitation Learning & Sensorimotor Feedback : https://github.com/JanhaviChaurasia/NeuroQuadruped
- Multi-Agent RL for Swarm Pursuit Coordination : https://github.com/JanhaviChaurasia/PacMan_MARL
- Multi-TurtleBot Coordination in Gazebo : https://github.com/JanhaviChaurasia/TurtleBot3_Swarm
- Perception for Autonomous Underwater Vehicle : https://github.com/Dreadnought-Robotics/Project-Mira